Engineering

1. **What is the function of the blower in a two-stroke diesel engine?**

a. To pump out the exhaust gases from the cylinder

b. To supply pressured air to the cylinder to burn the residual fuel

c. To supply pressured air to the cylinder to remove fuel fumes

d. To supply "fresh" pressured air to the cylinder to remove gases

2. What can happen when the valves' clearance is too large?

a. Noisy operation and decrease in output

b. Loss of compression and heating of the engine

c. Faster burning and increase in output

d. Easier start-up

3. What can insufficient small valves clearance cause?

a. Faster burning

b. Noisy operation

c. Burning of the exhaust valve

d. Burning of the suction valve

4. What closes the suction openings in a two-stroke petrol engine?

a. The piston

b. The valve

c. The cylinder

d. The sealing rings

5. What is the main function of the crankshaft counterweight?

a. Centrifugal balancing force

b. Weight increase of the crankshaft

c. Balancing force to the combustion pressure

d. Increase of the turning moment

6. What characterizes the fuel tubes from the injection pump to the spray-nozzles?

a. Straight and tight

b. Smooth in order to prevent pressure fall

c. They have same length

d. Temperature resistant

7. **What is the function of the fuel injector?**

a. To spray the oil into the fuel (mixture)

b. To spray the returned fuel

c. To spray the fuel into the injection pump

d. To spray the fuel into the combustion chamber

8. What will be the impact on the engine structure as the number of cylinders increase?

a. There is no impact due the number of cylinders, only the weight

b. The fly-wheel will become smaller proportionally to the number of cylinders

c. The fly-wheel will grow proportionally to the number of cylinders

d. The number of pistons will decrease proportionally

9. What is the main characteristic of a diesel engine?

a. An internal combustion engine and ignition by compression

b. An internal combustion engine and ignition by spark

c. An external combustion engine and ignition by compression

d. An external combustion engine of diesel

10. What is required prior to engine start-up?

a. Cleaning and oiling of gear systems

b. Connection to power and transmissions check-up

c. Set systems off and check up of liquids

d. Check for leaks and that the bearings are in order

11. What is required after start-up?

a. Check oil pressure and sea water disposal

b. Cleaning and oiling of systems

c. Check-up of liquids level as per need (oil, fuel, water)

d. Draining air and water

12. What are the functions of the piston rings?

a. Sealing the clearance between the piston and the cylinder

b. Stopping the piston movement prior to the end

c. Preventing the widening of piston

d. Regulation of the piston wear and tear

13. What is the main purpose of the fly-wheel?

a. Inertia preservation and rpm maintenance between strokes in the operation cycle

b. Balancing the turning movement of the starter

c. Providing momentum to the gear box

d. Maintaining regulated turning of the propeller

14. What are the main functions of the crankshaft?

a. Turning rotational movement arched

b. Turning linear movement to rotational movement

c. Acceleration of the pistons movement

d. Stopping the fast movement of the pistons

15. How is pressured lubrication (oiling) performed?

a. By the hydrostatic pressure of the oil

b. By the oil pressure pump

c. By the gravitational force of the oil

d. By centrifugal oil pressure

16. What is the operation order in a four-stroke diesel engine?

a. Suction, combustion, compression, exhaust

b. Compression, Combustion, suction, exhaust

c. Combustion, suction, compression, exhaust

d. Suction, compression, combustion, exhaust

17. Which mechanism performs (times) the valves operation?

a. The camshaft
b. The fly-wheel
c. The crankshaft
d. The timing chain

18. What are the pistons and connecting rods in the engine connected to?

a. To the crankshaft
b. Directly to the shaft
c. To the camshaft
d. To the fly-wheel through the connecting rod

19. How is the speed of the engine revolutions in a diesel engine adjusted?

a. By controlling the revolutions regulator
b. By the feeding pump
c. By controlling the air regulator
d. By the acceleration pump

20. Which cog-wheel does not turn during regular engine operation?

a. That of the timing shaft
b. That of the starter
c. That of the braking system
d. That of the ventilation mechanism

21. Which part in the fuel system determines the amount of fuel injected into the diesel engine?

a. The injection pump
b. The distributor
c. The supply (feeding) pump
d. The injection timer

22. In which area of the engine is the oil pressure measured?

a. In the oil suction line
b. After the oil filter
c. In the oil sump
d. Before the oil filter

23. What is the function of the piston?

a. To seal the cylinder space
b. To start the valves
c. To turn the burning pressure to linear movement
d. To suck the mixture

24. How is the fuel ignited in a diesel engine?

a. By the heat created due the compression of air
b. By an electric spark
c. By the heat created from the piston friction with the cylinder's side walls
d. By an electro-mechanic spark

25. What is the function of the connecting rod?

a. To restrain the pistons' shaking
b. To connect the piston with the crankshaft
c. To shake the camshaft
d. To connect the piston with the camshaft

26. What does the piston compress in a diesel engine?

a. Diesel drops with air
b. Air with fuel fumes (mixture)
c. Fuel fumes only
d. Air only

27. What is connected at both ends of the crankshaft?

a. The clutch at one end and the fuel and cooling pump on the other
b. The sea water pump at one end and the camshaft on the other
c. The propeller shaft directly at one end and the starter on the other
d. The gear box through the fly-wheel at one end and the wheel belt on the other

28. How is a diesel engine shut down routinely?

a. By stopping the fuel flow with the revolutions regulator
b. By disconnecting the electric power to the feeding pump
c. By disconnecting the air flow
d. By disconnecting the pumping system by mechanic command

29. Where is the oil in a diesel engine concentrated?

a. In the oil sump
b. In the pump and the piping system
c. In the upper absorption tank
d. In the cooling tank

30. What is the main function of the air blower in a two-stroke diesel engine?

a. To deliver more air into the cylinder in order increase the engine output
b. To fully utilize the temperature of the exhaust gases
c. To increase the engine's speed
d. To compress fresh air to the cylinder instead of the suction activity occurring in a four-stroke engine

31. What is the function of the turbo charger in a four-stroke diesel engine?

a. Reduction of the injected fuel amount and thus meaningful saving in fuel consumption
b. Increase of the engine's output
c. Increase of engine's revolutions speed by compressed air pressure into the cylinders
d. To fill-up more air which helps heating the fuel

32. On what does the output change in a diesel engine depend on?

a. The injection pressure
b. The type of injected fuel
c. The amount of injected fuel
d. The pump's speed

33. What is the basic purpose of the centrifugal pump in the engine?

a. Spiraling of the fuel
b. Injecting oil
c. Pumping water
d. Spiraling air

34. What is the function of the injection pump?

a. Providing fuel under pressure and timing to the spray-nozzles
b. Injecting oil to the pistons
c. Injecting water under pressure to the cooling system
d. Injecting excess fuel to the tank

35. What is the piston pump capable of?

a. Pumping water only
b. Pumping air and liquids
c. Pumping fuel only
d. The pump must have pressure in suction

36. What is the operation principle of the piston's pump?

a. Volume change
b. Rotational movement
c. Pressure change
d. Area change

37. What are the required positions of the automatic bilge pump's operating switch?

a. Switch-off, low output, high output
b. Switch-off, manual operation, automatic switch-on (float)
c. Switch-off, float operated, automatic switch-on
d. Switch-off, locked, automatic swith-on

38. What is the common failure in a sea-water pump?

a. Air penetration to the system
b. Blocking the suction opening
c. Blocking the exhaust opening
d. Lack of synchronization with the engine

39. What is the operating principle of the centrifugal pump?

a. The centrifugal speed turns to rotational movement
b. The liquid is thrown to the pump's center due to the rotational movement
c. The liquid is thrown to the pump's side walls due to the rotational movement
d. The centrifugal speed turns to force

40. What is the common use of the immersion pump?

a. Pumping of fresh water
b. Pumping immersed oil
c. Pumping fuel from the bottom of the tank in emergency
d. Pumping bilge water in an emergency

41. What is the common feeding pump in the engine's fuel system?

a. Diaphragmatic pump
b. Rotor pump
c. Impellor pump
d. Piston pump

42. On what principle does the rubber impellor pump (Jabsco) operate?

a. Pressure change
b. Speed change
c. Volume change
d. Centrifugal force

43. What can be the cause that the bilge centrifugal pump cannot dispose the water to the sea?

a. The centrifugal resistance is too low
b. The hydrostatic pressure at the exhaust is too low
c. The centrifugal load is too high
d. The hydrostatic pressure at the exhaust is too high

44. In a closed system, what is the function of the fresh (treated) water pump in the engine's cooling system?

a. To cool the sea water
b. To cool the oil
c. To bring the water back from the expansion tank
d. To transfer by the water the engine's heat in the heat exchange to the sea water

45. What is the function of the sea water pump in an ~~closed~~ open cooling system?

a. To spiral the water to cool the oil
b. To supply water to cool the freshwater system
c. To flow water to cool the incoming fuel
d. To supply water to directly cool the engine top

46. When the suction is unclogged, and the centrifugal pump cannot pump, what is the reason?

a. The rubber impellor is burnt
b. The safety valve was left in the 'open' position
c. Air is penetrating the suction line
d. The thermostat caused the pumping to stop

47. What is the function of the feed pump in the fuel system?

a. To feed the excess line back into the tank
b. Assistance in the air drainage (priming) and transfer fuel from the fuel tank to the injection pump
c. Assistance in feeding the fuel into the tank when vessel refueling
d. Performing the fuel spiraling between the fuel filters in order to pre-filter the fuel

48. Where is the feed pump in the fuel system generally located?

a. After the injection pump in the return line to the tank
b. Between the tank and the fuel filter
c. Between the oil filter and the injection pump
d. Inside the fuel tank

49. What transfers the main propulsion power of the propeller to the ship?

a. The tightening nut of the shaft
b. The conic end of the shaft
c. The groove in the shaft
d. The tightening screws of the shaft

50. What can a hole in the exhaust pipe of the engine cause?

a. Emission of gases into the ship's space
b. Emission of fuel drops and risk of fire
c. Increased air suction
d. Engine's overheating

51. What does "milky oil" (emulsion) in the engine signify?

a. Air in the oil
b. Fuel in the oil
c. Water in the oil
d. Calcium in the oil

52. The suction openings in a two-stroke petrol engine are closed by:

a. The piston
b. The valves
c. The liner
d. The sealing rings

53. What will be the impact if the propeller is changed to a larger pitch propeller?

a. The load on the engine will increase
b. The ship's speed will decrease
c. The load on the engine will decrease
d. There is no significance to the pitch but only to the diameter

54. What must be checked immediately after the engine's start-up?

a. Temperature, oil pressure, propeller's revolutions
b. Oil pressure, cooling water flow, charging current
c. Cooling water flow, oil flow, gases emission
d. Oil flow, charging current, vibrations

55. What are the essential spare parts of a diesel engine?

a. Spray-nozzles, belts, impellors, seals, filters
b. Propeller, relevant bearings, screws, seals
c. Pistons, seals, tubes, impellors
d. Valves, regulators, seals, relevant belts

56. What is the unit for measuring output?

a. Kilowatt/hour
b. Horse power per second
c. Watt
d. Meter per horse power

57. How is the change in a ship's direction from "ahead" to "astern" made?

a. Change in the rotation direction of the engine's shaft
b. Change in the rotation direction of the propeller shaft
c. Transmission change in the gear box
d. Crossing the operation belts

58. What is the liquid added to the battery?

a. Battery acid
b. Distilled acid
c. Diluted water
d. Distilled water

59. What is the purpose of the zinc fitted at the bottom of the ship?

a. Direct the water flow to the bottom
b. Prevent electric corrosion of metal parts in the water
c. Connect the electric grounding to the sea
d. Balance the propellers

60. What does the surface area of the ship’s propellor blades impact?

a. Power of the propulsion
b. Developing speed
c. Advance rate at every revolution
d. Shaft engine-revolutions ratio

61. How do you check whether fuel is reaching the injection pump?

a. Freeing the fuel tubes from the tank
b. Freeing the fuel meter
c. Opening the filter cap
d. Opening the air drainage at the pump's outlet

62. What will cause the operation of a diesel engine to turn irregular?

a. Penetration of air to the fuel
b. Loosening of the belt
c. Engine fatigue
d. Changes in the oil pressure

63. Why is the oil pressure high immediately after starting the engine?

a. There is residual oil in the system from the last operation
b. The oil level is still at the maximum
c. The oil is cold and more viscous
d. The oil pump has not yet reached its full speed

64. How does the electric ~~switch~~ starter turn the engine?

a. With the starting belt
b. With the cog-wheels
c. With the timing chain
d. With the starting shaft

65. What characterizes an integrated (closed) cooling system?

a. Engine is cooled by treated water cooled by sea water
b. Engine is cooled by sea water
c. Sea water are spiraled in a closed circuit
d. System is closed by a totally hermetic opening

66. How should fire in the engine chamber be extinguished?

a. Disconnecting fuel and air and extinguishing by ventilation
b. Disconnecting fuel and electricity, opening the chamber and putting out the fire with a fire extinguisher
c. Closing fuel, sealing ventilation openings to smother the fire and operating a fire extinguisher through a narrow aperture
d. Immediate flooding of the engine chamber with sea water

67. In which units is "output" measured?

a. Watt
b. Kg times meter/sec
c. Horse power
d. All the answers are correct

68. What is the volume of a cubic meter of water?

a. 10 liters
b. 1,000 liters
c. 100 liters
d. 10,000 liters

69. In which type of cooling system will heat sink and a radiator appear?

a. Closed
b. Air
c. Open
d. Water

70. A quarter cubic meter of diesel oil left is in the tank. The engine consumes 8 liters of fuel per hour. How long approximately can the engine be operated?

a. 31 hours
b. 18 hours
c. 21 hours
d. 42 hours

71. Is it necessary to operate the engine in order to engage the electric anchor windlass?

a. No, it can only hamper
b. It is desirable, in order to assist in the current supply
c. No, the anchor's rope can be entangled in the propeller
d. Yes, in order to tighten the anchor chain with the propeller

72. What could the sediments in the fuel system be?

a. Heavy oil particles
b. Solid particles
c. Heavy fuel particles
d. Heavy chemical particles

73. When will the oil pressure be higher?

a. When the engine is started cold
b. When the engine is operating hot
c. Prior to start-up
d. In the last start-up (the oil is more viscous)

74. Where is the air filter located in a diesel engine?

a. Behind the air pump
b. Inside the air pump
c. On the evaporator
d. On the suction manifold

75. What is the main purpose of the oil in the engine?

a. Increase fuel viscosity
b. Improve the sealing between the piston and the engine top
c. Wet the parts and avoid heating
d. Decrease wear and tear and friction

76. What is the phenomenon caused in the compression stroke of a diesel engine?

a. Increase of pressure and temperature of the mixture
b. Improved sealing between the piston and the engine top
c. Increase of air pressure and temperature
d. Reduction of wear and tear and friction

77. What is the function of the fuel nozzle spray in a diesel engine?

a. Dissolve the fuel into aerosol and spray it in the combustion space
b. Supply the cylinders pressurized fuel
c. Supply the pump fuel particles
d. Determine the amount of sprayed fuel

78. What is the function of the revolutions regulator in a diesel engine?

a. Regulate and stabilize the pistons operation
b. Stabilize the operation of the feed pump
c. Stabilize and time the operation of the fuel nozzle sprays
d. Keep the speed of the engine revolutions desirable and steady

79. What is the main purpose of the fuel system in a diesel engine?

a. Vaporize the fuel timely and in the correct amount
b. Spray fuel into the combustion chamber on time and in the correct quantity
c. Create a precise mixture of fuel and air
d. Create the precise mixture of fuel and oil

80. How can the amount of oil in the sump be checked?

a. By the oil level meter
b. By the dip stick
c. By the oil pressure meter
d. By a transparent tube (connected vessels)

81. What is the electric current produced by the battery?

a. Direct single-phase current
b. Alternating single-phase current
c. Pulsating direct current
d. Regular direct current

82. How can the positive and negative polarity in the battery be determined?

a. The positive pole is thinner
b. The positive pole is always on the right hand side
c. The positive pole is always thicker
d. The positive pole is always colored red

83. What is the hydrometer used for?

a. The density of the solution in the battery
b. The amount of water
c. The acid temperature in the battery
d. The battery voltage

84. If it is not marked, how can it be known what voltage the battery can produce?

a. With an ammeter
b. By the number of cells
c. With a micrometer
d. By the size of the battery

85. What protects the electric system from overload?

a. The breaker
b. The distributor
c. The regulator
d. The fuse

86. What should be done when the voltage produced by the alternator is not steady?

a. Disconnect the alternator
b. Check the ~~starting~~ alternator belt ~~of the alternator~~
c. Increase the engine's revolutions
d. Disconnect the battery

87. In what units is the battery capacity measured?

a. Watt/hour
b. Volt
c. Ampere/hour
d. Kilowatt/hour

88. What is the function of the battery in the ship?

a. Turn thermal energy into electric energy
b. Turn chemical energy into electric energy
c. Provide electric energy to the engine
d. Provide electric energy to operate electric systems

89. During daytime, when the engine is operating without the use of electric consumers, the ammeter shows 5 ampere for a long duration. What is the significance?

a. It does not matter as long as the instruments work properly
b. There could be leakage/short circuit in the electric system
c. The battery is properly charged
d. The battery is full

90. During operation and transmission by communication instruments the voltage meter drops from 13.5 volt to 12.5 volt. What is the significance?

a. The alternator is not properly charged
b. The contact on the communication instrument is loose
c. The batteries are ok
d. The voltage regulator is out of order

91. What is the purpose of the "clearance" between the valves and the rocker arms in the engine?

a. To prevent over-friction of rocker arms by the burning gases
b. To enable the cooling of the suction valves
c. To enable thermal expansion of the valves' operating mechanisms
d. To regulate the flow of the burning gases and the fuel mixture

92. The temperature is 176 degrees Fahrenheit, what is the approximate temperature in degrees Celsius?

a. 125 degrees C
b. 80 degrees C
c. 45 degrees C
d. 195 degrees C

93. When is the risk of explosion in the fuel tank greater?

a. When the fuel tank is almost empty
b. When the tank is half full
c. When the tank is full
d. All the answers are correct

94. What are the sediments in the fuel tank of a diesel engine likely to be?

a. Heavy oil particles
b. Bacteria and mushrooms developed in the fuel tank
c. Fuel particles
d. Heavy soot particles

95. The thermometer in a diesel engine shows 213 degrees Fahrenheit, what is required to be done?

a. Engine is in boiling stage and should be shut down and cause checked
b. Engine should be immediately accelerated and pressure of the cooling water increased
c. Everything ok, nothing should be done
d. Replace the initial fuel filter

96. The thermometer of the cooling water in the engine shows 180 degrees C, what is the significance?

a. The temperature is too high
b. The temperature is reasonable
c. The temperature is too low
d. The thermometer is probably out of order

97. What is the main purpose of the "kyle" at the end of the propeller shaft?

a. Used as the main power transfer to the propeller
b. Used as a centrifugal balancing force
c. Used for the locking and strengthening control of the propeller to the shaft
d. To increase the turning moment

98. How are the engine parts of a petrol two-stroke engine oiled?

a. By the oil mixed with fuel
b. By spraying the oil in the sump
c. With the assistance of a mechanical oil pump
d. By injection from an external tank

99. What generally characterizes the tubes transporting fuel from the injection pump to the spray-nozzles?

a. They are straight and tight
b. They are usually the same length and high-pressure resistant
c. They are made of steel, smooth and flat in order prevent pressure fall
d. They are temperature resistant up to 900 degrees Celsius

100. What is the oil color indicating that water penetrated the engine's oil?

a. Reddish
b. Purplish
c. Whitish
d. Blackish

101. What can an increase in the level of oil in the sump indicate?

a. Penetration of air to the oil
b. Penetration of fuel to the oil
c. On a hole in the exhaust manifold
d. Prolonged use of the oil

102. What is the main function of the gear box in most ships?

a. To Adapt the engine to the existing load
b. To enable change of the turning direction of the engine
c. To enable change in the speed of the shaft
d. To enable change of the turning direction of the propeller shaft

103. Where does the starter get the electric power for its activation from?

a. From the dynamo
b. From the alternator
c. From the battery
d. From the generator

104. What is the function of the thrust bearing in the propeller shaft system?

a. Create pressure on the gear box due to the propeller activity
b. Transfer of the propulsion force from the propeller to the ship
c. Decrease of the engine pressure on the gear box
d. Curb the centrifugal force of the propeller

105. After sailing under load and during the mooring to the wharf, after reduction of the engine revolutions, the oil pressure alert is on, what is the required response?

a. The engine must be shut down immediately, otherwise it will be completely destroyed
b. The revolutions should be increased, in order for the oil pressure to increase again
c. The mooring has to be quickly completed and the engine shut down
d. The mooring should be discontinued, the port i to be left again and the failure to be checked

106. What is the purpose of the ventilation pipe located on the fuel tank?

a. To enable freeing and disposal of water vapor to the atmosphere
b. To enable breathing of the fuel tank and equalize the pressure in the tank to the atmospheric pressure
c. To enable cleaning of the tank from sediments
d. All the answers are correct

107. What is the principal difference between a petrol and a diesel engine?

a. The type of fuel
b. The compression ratio
c. The fuel ignition method
d. All the answers are correct

108. How does the fuel in a diesel engine ignite?

a. The fuel is heated before being injected into the cylinder
b. By an electric plug
c. It ignites due to contact with the compressed air and heat in the cylinder
d. No answer is correct

109. What will happen if we activate a Jabsco pump when its suction valve is closed?

a. The sub-pressure in suction will reach maximum and the pump will disintegrate
b. The impellor will burn and be damaged
c. The safety valve of the pump will open
d. No answer is correct

110. What is the purpose of the zinc fingers in the sea water cooling system?

a. To prevent the clogging of the filters and pipes with salt sediments
b. To prevent rusting in the coolers and pipes
c. To prevent corrosion in the parts of the sea water system
d. All answers are correct

111. If the feed pump of the diaphragm type is not pumping, what can be the possible cause?

a. The activating cog of the pump is at its peak and the engine should be turned half a revolution
b. The diaphragm is torn or the pump's valves are defective
c. The net filter in the pump is clogged
d. All answers are correct

112. What is the purpose of the valve in the cap of the expansion tank?

a. To seal the cover against leaks
b. To keep high water pressure in the system and increase its boiling point
c. To decrease the water pressure in the system and lower its boiling point
d. Answers a and c are correct

113. While performing the engine oil check it is discovered that its color is black, what must be done?

a. The oil has to be replaced immediately because it points to a failure
b. There is no need to do anything. The oil must absorb soot particles as well.
c. The oil does not fit the engine and must be replaced with another type
d. The oil filter must be urgently replaced with a new one

114. What is the meaning of "over charge"? (turbo charge)

a. Bringing in larger volume of air to the cylinder of a given volume
b. Utilization of the burning gases to bring in higher volume of air to a cylinder of a given volume
c. Bringing in higher weight of air to a cylinder of a given volume
d. Recycling the pressure of the burning gases exhausted from the cylinder

115. Why does every engine lose lubricating oil while operating?

a. Because the oil in the sump dissipates even in ambient temperature
b. Because part of the oil lubricating the piston rings is eventually burned
c. Because there is no engine, even the newest one, that has no oil leaks
d. All the answers are correct

116. Why it is prohibited to smoke near an electric battery being charged?

a. The smoke is harmful to the battery
b. The smoke and gases mixture released from the battery while charging are harmful to the charger
c. The hydrogen emitted during charging could explode
d. All the answers are correct

117. What must be checked during the battery charge?

a. That the ventilation nozzles in the caps are open
b. That the battery is disconnected from the electrical system
c. That the ventilation caps are closed
d. That the battery will not supply simultaneously voltage to consumers

118. During the start-up attempt of a diesel engine, the engine is rotating well but not started, what could be the cause?

a. The oil pressure is low
b. The electric contacts in the relay of the starter are out of order
c. No fuel is arriving to the spray nozzles
d. All the answers are correct

119. You have started a diesel engine and after few minutes the engine shut itself, what could be the cause?

a. You forgot to connect the main electric switch
b. The alternator whose function is to charge the battery is out of order
c. Lack of fuel or failure in the fuel system
d. The battery is depleted

120. Why is operating the engine in low speed for a prolonged time undesirable?

a. Air could penetrate the fuel system
c. Soot could accumulate on the piston, the valves and at the end of the spray nozzle

b. The engine is not getting sufficient cooling because the water pumps are revolving slowly

d. All answers are correct

121. Which signs could indicate that the engine is operating on overload?

a. Cooling water temperature is too low and black smoke at exhaust

b. Cooling water temperature too high and black smoke at exhaust

c. The engine revolutions cannot be increased

d. Answers a and c are correct.

122. While the engine is operating, it starts to "stutter", what could be the cause?

a. The sea water suction valve was erroneously closed

b. Probably the fuel tank became empty, and the feed pump sucks air as well

c. The engine is straining because of overload

d. No air drainage was performed on the fuel system as should be done prior to every start up

123. While sailing, the fuel in the tank ran out and the engine shut down, what is the next step?

a. Start up the engine again and check pressures and temperatures

b. Clean the secondary fuel filter

c. Fill the fuel tank and perform air drainage to the fuel system

d. Start up again and check whether the feed pump is disposing overboard

124. 5 Cubic meters of water are equal in volume to:

a. 5000 gallons

b. 500 liters

c. 50 liter

d. 5000 liters

125. Output of 250 horse power is approximately equal to:

a. 175 kilowatt

b. 186 kilowatt

c. 236 kilowatt

d. 265 kilowatt

126. When should the oil level of the engine be checked?

a. Prior to engine start up

b. After engine start up

c. While the engine is operating

d. It has no importance as in every stage the measurement is identical

127. What does the thermometer in a diesel engine generally measure?

a. The diesel temperature in the excess line

b. The oil temperature in the sump

c. The water temperature in the oil system

d. The water temperature coming out of the ~~engine top~~ cylinder head

128. How is a diesel engine routinely shut?

a. Disconnecting the fuel flow with the control rod attached to the revolutions regulator

b. Disconnecting the electric power to the feed pump

c. Disconnecting the air supply

d. Disconnecting the pumping mechanism by a mechanical control command

129. What can an engine’s loose belt cause?

a. Off-key engine operation
c. Deficient batteries charge

b. Decrease of oil pressure
d. Decrease in the fuel flow rate

130. What will be the effect on the ship if we change the propeller to one of the same diameter but smaller pitch?

a. The load on the engine will increase
c. The load on the engine will decrease

b. The vessel speed will increase
d. There is no significance to the pitch but the diameter only

131. What does the propeller pitch indicate?

a. The force intensity in every revolution
c. The advance of the vessel in every revolution

b. The pressure intensity in every revolution
d. The water pressure resistance braking the vessel

132. What is the main impact emanating from the size of the propeller diameter?

a. The propulsion force transferred to the ship
c. The vertical pressure in every revolution

b. The vessels' advance in every revolution
d. The required shaft length size

133. Why is the diameter of the piston smaller than the diameter of the cylinder?

a. To enable thermal expansion
c. To enable the passage of air

b. To create pressure
d. There is no clearance between the cylinder and the piston

134. What is the name of the lowest point the piston reaches while working?

a. The bottom point
c. Upper dead point

b. Lower dead point
d. Piston minimum

135. How is the gear box oiled?

a. By the gear box oil
c. There is no oil in the gear box

b. By the water flowing through it
d. From the oil system of the engine

136. What is the purpose of the oil in the engine?

a. To cool the engine
c. To clean the soot

b. To oil the engine parts
d. All answers are correct

137. What does the shut down handle control?

a. The fuel tap of the tank
c. Closure of the air to the engine

b. The revolutions regulator
d. The feed pump

138. What does the throttle control?

a. The gear
c. The revolutions regulator

b. The battery
d. Answers a and c are correct

139. Which indicators must be on the control panel of the engine?

a. Oil pressure, engine temperature, engine rpm
b. Just the oil pressure
c. Fuel quantity, exhaust gases temperature, engine heat
d. Oil pressure, engine rpm, gear position

140. What is the purpose of the injection pump?

a. To provide high pressure fuel to the spray nozzles, in varying amounts and precise timing
b. To provide fuel to the spray nozzles according to the injection order and in the shape of tiny drops
c. To change the fuel pressure and maintain engine's fixed and steady speed
d. To supply fuel in the shape of tiny drops to the spray nozzles

141. What are the basic conditions required for catching fire?

a. Inflammable material, oxygen, heat
b. Vapor, temperature, air
c. Liquid, oxygen, temperature
d. Inflammable material, temperature, pressure

142. What must be checked prior to engine start up?

a. Fuel quantity, engine idle, sea water outlet and oil pressure
b. Oil level, engine idle, electricity and air
c. Engine idle, temperature, oil pressure, fuel quantity
d. Oil level, fuel quantity, quantity of cooling water, idle engine

143. What actions should be taken when the engine is overheating?

a. Reduction of load and locating the failure
b. Adding water
c. Cooling the engine
d. Checking the sea water level

144. What can cause a significant reduction and rapid loss in the amount of the cooling liquid in a closed cooling system?

a. Cooling of the engine and water shrinkage
b. Engine overheating and boiling
c. Revolutions speed higher than normal
d. Leaks in the open cooling system

145. What is the apparatus that causes the closure of the air entry openings in a two-stroke petrol engine?

a. The piston
b. The liner
c. The valves
d. The sealing rings

146. What is the common failure in diesel engines?

a. Penetration of air to the oil system
b. Penetration of air to the fuel system
c. Penetration of water to the fuel system
d. Penetration of water to the air system

147. What can cause the "diluted oil" phenomenon?

a. Mixing air with oil
b. Penetration of water to the oil
c. Penetration of fuel to the oil
d. Penetration of cold combustion gases to the oil

148. What protects the electric device from over-current or electric short circuit?

a. The battery
b. The alternator
c. Shore power switch
d. The fuse

149. What are the phenomena resulting from a dirty oil filter?

a. Decrease of oil pressure
b. Increase of oil pressure (no outlet))
c. Decrease of compression (no oil)
d. Increase of compression

150. What causes the ignition of fuel in the combustion chamber of a diesel engine?

a. A high voltage spark
b. The extremely hot compressed air in the cylinder
c. A chemical ignition mechanism
d. A turbo electric ignition mechanism

151. How will you act in the case of a low oil pressure warning?

a. Immediately check the oil meter reading
b. Immediately shut down the engine
c. Increase the speed of revolutions (increase of pressure)
d. Reduce the speed of revolutions (cooling of oil)

152. How the main bending moment of the propeller to the shaft is transferred?

a. By the tightening nut at the end of the shaft
b. By the "kyle" in the shaft
c. By the conic end of the shaft
d. By the tightening bolts of the propeller

153. What causes the fuel ignition in a two-stroke petrol engine?

a. An electric spark plug
b. An electrochemical spark plug
c. The pressure and heat created in the combustion chamber
d. The pressure and heat of the mixture in the compression activity

154. What is the purpose of the breather pipe, located in the fuel tank?

a. To let in cold air to the tank in order to prevent explosion
b. To release water vapor into the atmosphere
c. To enable the fuel tank "to breathe and compensate by air" while refueling or pumping fuel
d. To take out the excess fuel from the fuel tank through the excess line

155. What is the purpose of the thermostat fitted in the cooling system of the engine?

a. To enable engine operation in a stable and fixed temperature
b. To enable rapid cooling of the engine
c. To keep the temperature of the outgoing fuel
d. To keep the temperature of the incoming fuel

156. The thermostat in the cooling system is stuck in the open position, what could be the consequences?

a. There is no affect whatsoever to that
c. In the long run there will be damage, because the engine will work at a temperature lower than recommended

b. The engine could boil as a result

d. The pistons might get stuck due to unstable temperatures

157. At what frequency is it recommended to check the cooling water level?

a. Once a day

b. After every engine operation

c. Prior to every engine start up

d. At least once a week

158. A heat exchanger (cooler) is characteristic to the following cooling unit type:

a. Closed cooling system

b. Open cooling unit

c. Air cooling unit

d. Water cooling unit by radiation

159. What is the operating principle of a petrol engine?

a. Internal combustion and ignition by compression

b. External combustion and ignition by compression

c. Internal combustion and ignition by spark

d. External combustion of diesel

160. When will the oil pressure be lower than usual?

a. When there is lack of oil in the system

b. After prolonged activity under load and heating of the engine above normal heat

c. When the engine is started cold

d. When there is pressured air penetration into the oil system

161. At start up, the diesel engine is revolving and does not manage to start, what could be the cause?

a. There is air in the fuel system

b. Electrical failure (no ignition)

c. There is a fuel leak

d. The oil pressure is low

162. Which steps should be taken after discovering air in the fuel system?

a. Suction of the air from the fuel

b. Shut down and restart

c. Drainage of the air from the fuel system

d. Adding fuel to the system (to prevent air entry)

163. Which steps should be taken when the revolutions of a diesel engine increase beyond control?

a. Immediately shut down the engine by smothering the air

b. Adjustment of the fuel regulator

c. Immediate shutting off the fuel tap on the tank

d. Adjustment of the air regulator

164. What is the function of the revolutions regulator in a diesel engine?

a. Issues commands to the throttle to keep desirable engine speed

b. Keeps desirable engine speed (rpm), by issuing commands to the injection pump

c. Issues commands to the feed pump, to change the amount of the injected fuel

d. Keeps desirable engine speed, by issuing commands to the spray nozzles

165. Where is the thermostat in a cooling unit of the engine usually located?

a. At the outlet of the cooling water, from the ~~engine top~~ cylinder head

c. Fitted at the outlet opening of the sea water pump

b. Fitted at the entrance of the cooling water to the cylinder block

d. No answer is correct

166. The engine operates OK in idle position, but shuts down when engaged forward, what do you think happened?

a. A rope or nylon were caught by the propeller

b. There is excess oil in the gear box

c. Air has penetrated the fuel system

d. Water has penetrated the fuel system

167. What is required to be done routinely in order to shut down a diesel engine?

a. The throttle rod must be moved to "middle" position, otherwise the engine will not shut down

b. Disconnecting the fuel to the spray nozzles by pulling the shut down handle connected to the fuel regulator

c. Closing the fuel supply valve on the tank

d. Moving the throttle backwards till the end and pulling the shut down handle

168. What are the signs that may point to lack of lining of the propeller shaft?

a. Vibrations of the ship's hull

b. Vibrations of the engine and gear box

c. Too large water leak from the gland

d. All the answers are correct

169. What must be done to perform "priming" (air drainage) to the tubes of the spray nozzles?

a. The engine should be continuously operated (the crankshaft) and the tubes connectors to the spray nozzles be released

b.The feed pump should be operated manually

c. The throttle (fuel throttle) should be moved to engine shut down position

d. Answers a and c are correct

170. What is the main function of the cooling pump in the fresh water system?

a. To supply water to cool the oil gear box cooler

b. To supply water to cool the sea water cooler

c. To pump water from the expansion tank, located at the highest point in the system

d. No answer is correct

171. What is the purpose of the lubricating oil in the engine?

a. To reduce friction between the moving parts

b. To assist in preventing rust

c. To cool the bearings, the pistons and other moving parts

d. All answers are correct

172. What is the purpose of the gland in the shaft propeller system?

a. To carry the weight of the propeller shaft at the point it exits the vessel to the sea

b. To prevent leaking of dirty bilge water from the vessel to the sea

c. To prevent penetration of sea water through the exit opening of the propeller shaft to the sea

d. No answer is correct

173. In which units is the battery capacity measured?

a. Volt

c. Watt-hour

b. Ampere

d. Ampere-hour

174. What is the significance when operating the navigation lights, that the voltage measured on the battery poles is 10 volts

a. The electrical connections of the communication instruments are out of order

b. It is a natural situation, the battery is ok

c. The communication instrument operates on 6 volts and therefore there is no problem

d. The battery is weak and probably out of order

175. What is one of the required conditions to start a diesel engine?

a. Hot air must be injected at high pressure, which will ignite the fuel in the cylinder

b. It is enough to inject fuel to the cylinders in order to start the engine

c. The crankshaft must be rotated fast in some way

d. No answer is correct

176. At start-up a knock is heard from the starter, however the engine does not turn, What could be the cause?

a. The battery is weak

b. The engine is stuck and cannot turn

c. The electrical contact of the solenoid (relay) of the starter are out of order

d. All the answers are correct

177. At start-up, the engine turns too slow and is not "started", what could be a probable cause?

a. Oil pressure is too low

b. The battery triggers are not properly connected, or the battery is weak

c. The shut down handle is in shut down position

d. Failure of the start-up switch

178. You have started the engine and discover that water is not coming out of the exhaust pipe, what could be the cause?

a. The sea water entry valve is in a closed position

b. Lack of water in the expansion tank

c. No water should be coming out of the exhaust pipe, only gases

d. You forgot to check the sea water level, prior start up